Computer Architecture

Project Name: Logic Gates Tester
Topic: Logic gates
Group: 03
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Introduction to Logic Gates:

Logic gates are the basic building blocks of digital electronics. They take binary input (0 and 1) and produce an output (0 or 1)

Seven Common Logic gates:

1. AND gates:

AND gate gives output 1 only when both inputs are 1. otherwise the output is 0.

2. OR Gate:

ORgate gives output 1 when at least 1 input is 1. It gives output 0 when both inputs are 0.

3. NOT Gate:

NOT gate changes the input into its opposite value. If input is 0 output is 1. If input is 1, output is 0.

4. NAND Gate:

It is the opposite of AND Gate. It gives output 0 only when both inputs are 1.

5. NOR gate: It is the opposite of the OR gate. It gives 1 only when both inputs are 0.

6. XOR gate:

It gives output 1 when both inputs are different. If both same output 0.

7. XNOR:

It gives output 1 when both inputs are same If different output 0.

Truth Tables.

AND Gate:

A	B	output
0	0	0
0	1	0
1	0	0
1	1	1

NOT Gate

A	B	output
0	#	1
1	"	0

OR Gate:-

A	B	output
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate:

A	B	
0	0	1
0	1	1
1	0	1
1	1	0

CamScanner

NOR Gate

A	B	output
0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate:

A	B	output
0	0	0
0	1	1
1	0	1
1	1	0

XNOR Gate.

A	B	output
0	0	1
0	1	0
1	0	0
1	1	1

Solved Examples:.

AND Gate.

Input A = 1
" B = 1

Operation 1 AND 1

output = 1.

XOR Gate.

Input A = 1
" B = 0

Operation = 1 XOR 0

Output = 1.

1. START
2. Select a logic gate.
3. Get input A.
4. Get input B (except for NOT gate.
5. Check which gate is selected.
6. Perform selected gate operation.
7. Calculate the output.
8. Display the output.
9. Display truth table of the selected gate.
10. END.